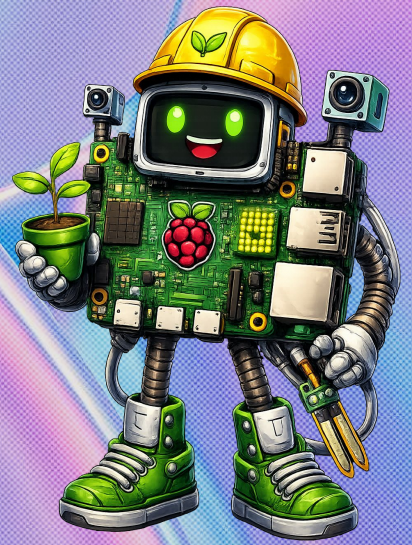
FloraGuard Hackathon 2026



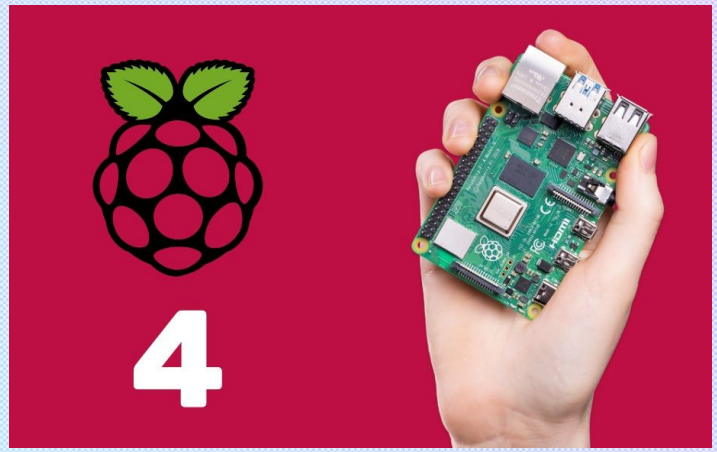
A Smart Plant Monitoring System

By Hurrain, Irene, Izabela

Introduction

FloraGuard

- FloraGuard is a Smart Plant Monitoring System.
- Using intelligent sensors and data analysis, FloraGuard monitors plant conditions in real time, giving you insights into the environment and how it affects your plants.
- The system uses a **Raspberry Pi** and **Sense Hat** to collect and process data, making it easy to understand and act on the information to help your plants thrive.



01

Project Overview

User Alerts - Email sent when change occurs

rubuslabs@gmail.com
to me

🌿 FloraGuard has detected environmental issues.

Current Readings:
Temperature: 24.71 °C
Humidity: 22.51 %

•••

Light: 2.00 Lux

⚠️ Issues detected:
TOO DRY

💡 Recommended Actions:
• Follow FloraGuard guidance for the above issues

This is an automated alert from FloraGuard.

Stay green 🌱
Rubus Labs | FloraGuard

Reply Forward

rubuslabs@gmail.com
to me

🌿 FloraGuard has detected environmental issues.

Current Readings:
Temperature: 21.91 °C
Humidity: 30.94 %
Light: 0.00 Lux

⚠️ Issues detected:
LOW LIGHT

•••

💡 Recommended Actions:
• Follow FloraGuard guidance for the above issues

This is an automated alert from FloraGuard.

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Reply Forward

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🌿 FloraGuard has detected environmental issues.

Current Readings:
Temperature: 32.38 °C
Humidity: 33.20 %
Light: 1.67 Lux

⚠️ Issues detected:
TOO HOT, LOW LIGHT

•••

💡 Recommended Actions:
• Follow FloraGuard guidance for the above issues

This is an automated alert from FloraGuard.

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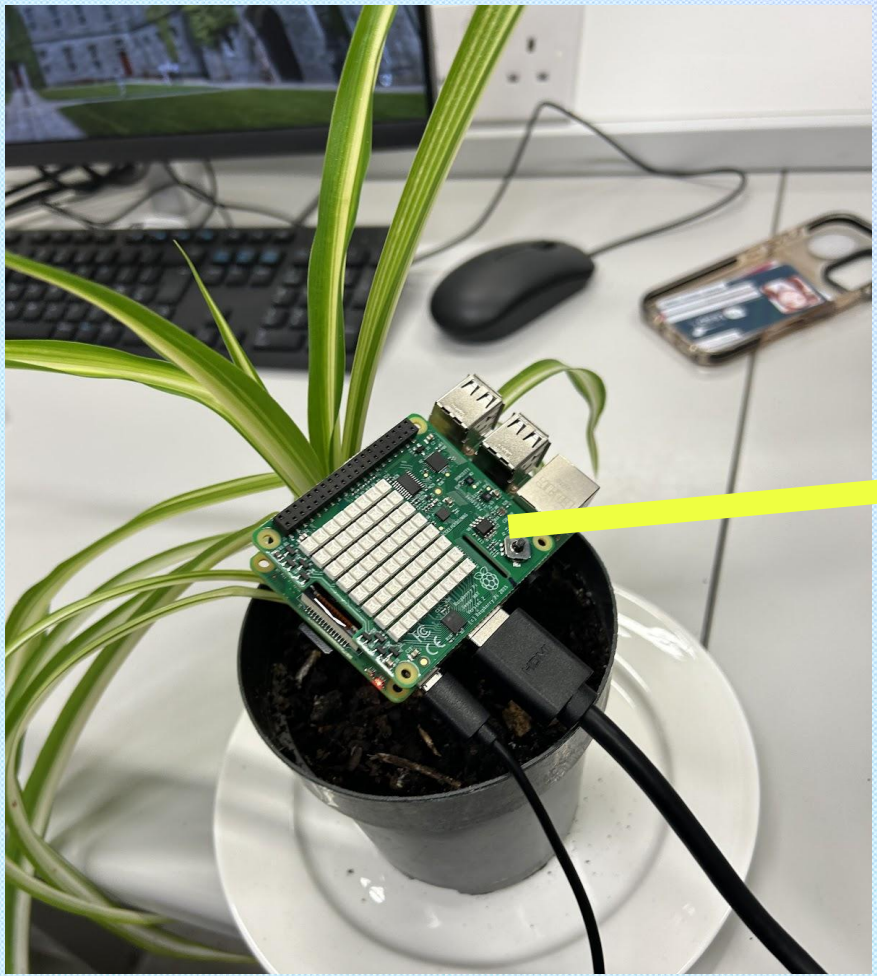
Reply Forward

Problem Statement: Monitoring Plant Health Challenges

- Plants require precise environmental conditions for optimal growth.
- Monitoring these variables manually is inefficient and prone to error.
- Challenges include detecting soil moisture levels, sunlight exposure, and ambient temperature variations. Without continuous monitoring, plant care may suffer, leading to poor health or crop failure.

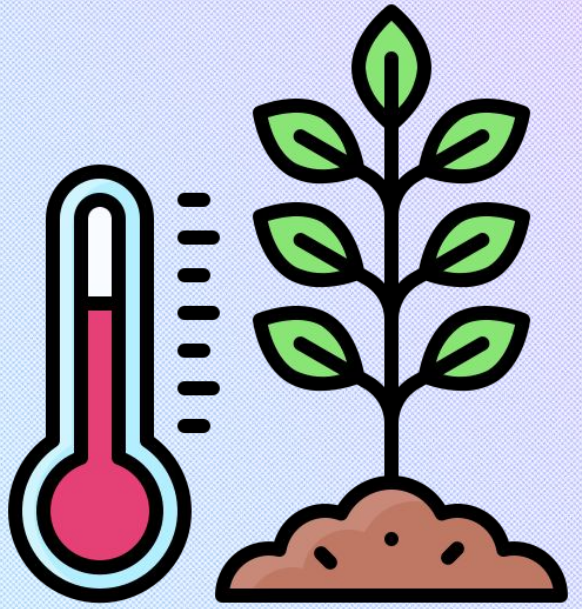


Sense Hat

Proposed Solution: FloraGuard System on Raspberry Pi

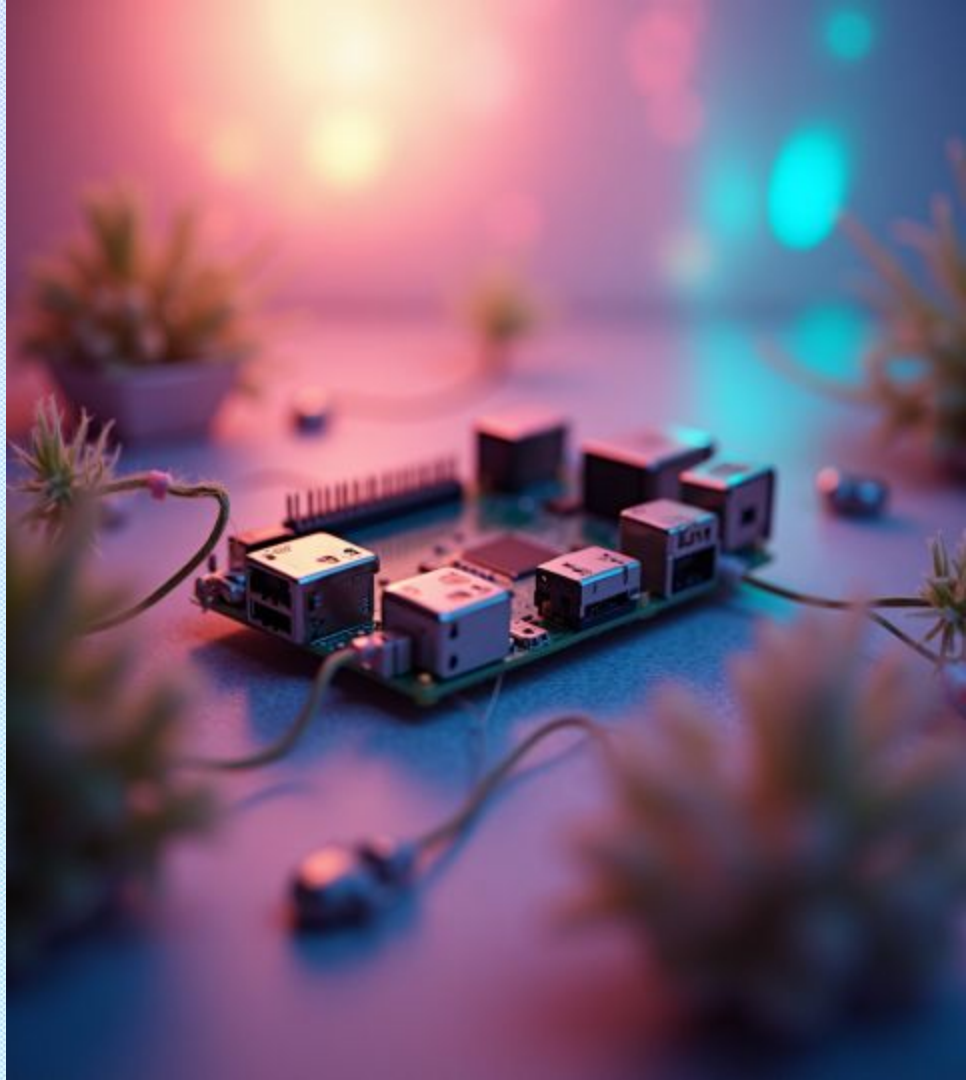
- FloraGuard integrates the Raspberry Pi with the Sense HAT sensors to continuously monitor environmental parameters including temperature, humidity, and soil moisture.

- Data is analysed locally and reported to users via a user interface. Alerts and recommendations enable timely interventions, improving plant care and reducing resource waste.

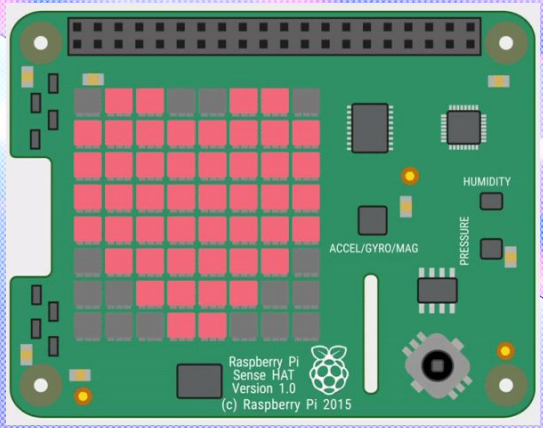


02

Technical Components



Data Analysis, Feedback, and Logging

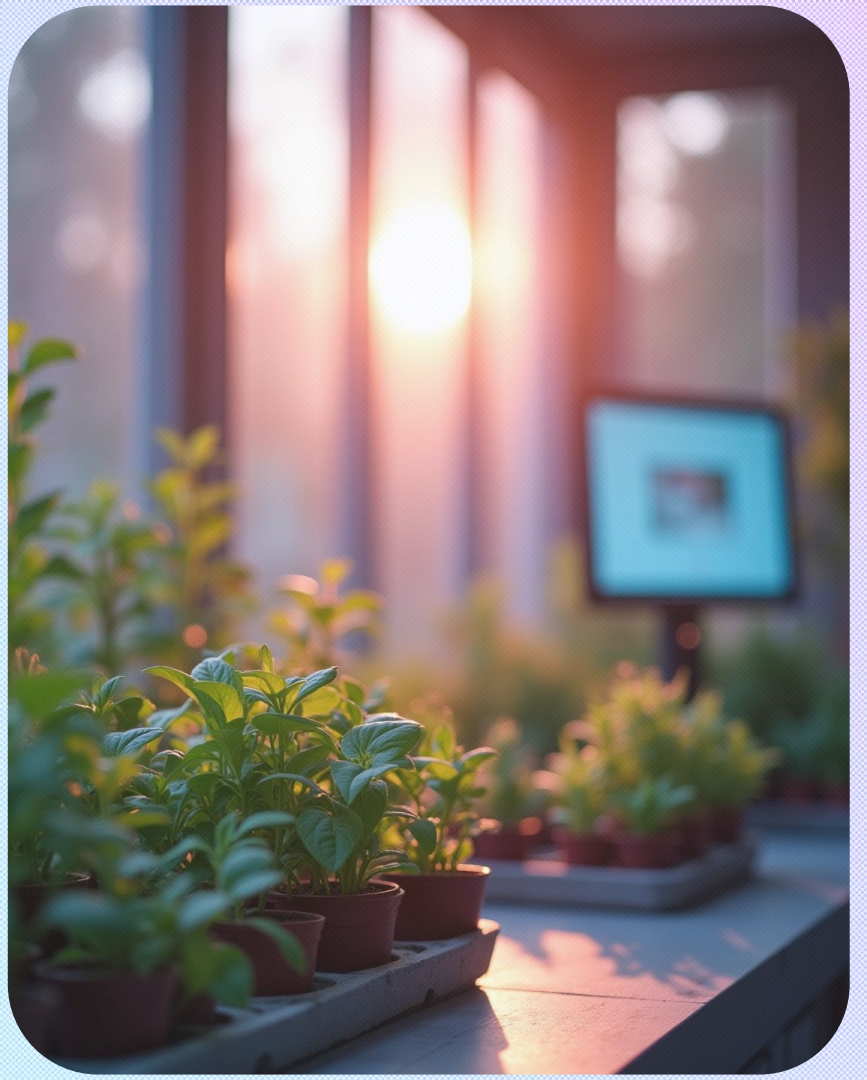


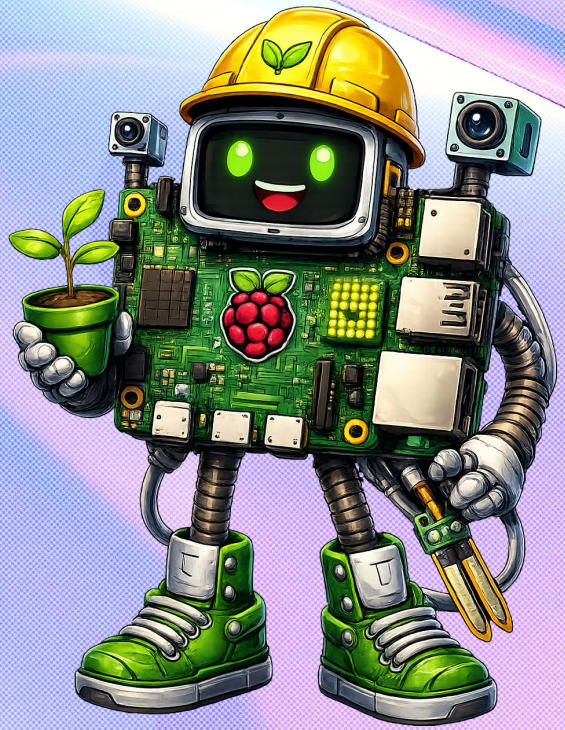
- The system uses Python scripts to analyse live sensor data, comparing it to optimal thresholds for different plant species.
- An 8x8 LED matrix provides instant visual feedback, showing plant health through easy-to-understand, color-coded icons.
- On top of that, hourly readings are saved in CSV files on the Raspberry Pi, allowing for long-term tracking and the creation of growth trend reports.

Timestamp	Temperature (°C)	Humidity (%)
2026-02-28 16:38:16	15.19	35.66
2026-02-28 16:43:18	14.45	51.44
2026-02-28 16:43:18	23.33	75.86
2026-02-28 16:43:24	14.41	34.88
2026-02-28 16:47:31	26.74	31.00
2026-02-28 16:47:35	25.02	66.08
2026-02-28 16:47:35	29.95	66.84
2026-02-28 16:47:35	14.01	77.16
2026-02-28 16:52:37	7.79	70.80
2026-02-28 16:52:41	8.35	42.72

Conclusions

FloraGuard brings together smart sensors and intelligent software to provide a complete plant monitoring solution. By continuously collecting and analysing environmental data, it gives users real-time feedback and long-term insights into plant health. The addition of visual monitoring adds an extra layer of observation, making FloraGuard a valuable tool for both indoor gardeners and larger-scale agricultural applications.



End of Presentation

Thank you!